TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Software for Yarn Manufacturing

The monitoring and management software of a spinning mill: how every frame reports its production, ends-down and stoppages to one screen — and how the shift report that management actually reads is calculated from raw machine data.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 5 · LEVEL 3 TERM 2

1. Aim

To use yarn-manufacturing (spinning mill) software to build a production-monitoring record for a ring-frame section: entering machine and count data, computing theoretical production and efficiency, analysing ends-down, and producing the shift report with its exception flags.

2. Concepts you need

- **What the software watches:** a mill monitoring system collects data from sensors and machine controllers across blowroom, carding, draw frame, comber, simplex, ring frame and winding. The core quantities per machine and shift are: production (kg or hanks), running time vs stoppage time with reason codes, **efficiency**, and for ring frames the **ends-down** (yarn breakage) count.
- **Theoretical (100 %) production** is pure arithmetic from machine settings — spindle speed, twist, count and spindle number. **Efficiency = (actual production ÷ theoretical production) × 100**. The gap between the two is where stoppages, ends-down and slow-running hide; monitoring software exists to make that gap visible machine by machine.
- **Ends-down rate** is stated per 1,000 spindle-hours so frames of different sizes compare fairly: *rate = breaks ÷ (spindles × running hours) × 1,000*. A rising rate on one frame points to that frame (or its roving supply); a rise on all frames points to the material or the humidity.
- **Downtime reason codes:** when a machine stops, the system logs the reason — doffing, ends-down cluster, mechanical, electrical, no material, power. The shift's lost production is then attributable, not anecdotal.
- **Laboratory link:** the same software family holds the quality record — count CV%, U%, strength results entered from the QC lab (TE-3201/TE-3202 work) against each production lot, so a buyer's lot can be traced from bale to cone.

- **Reports, not screens, run the mill:** the shift report (production, efficiency by machine, worst three frames, ends-down exceptions), the count-change plan and the maintenance alert list are the outputs people act on. Software that cannot produce them in one page has failed its purpose.

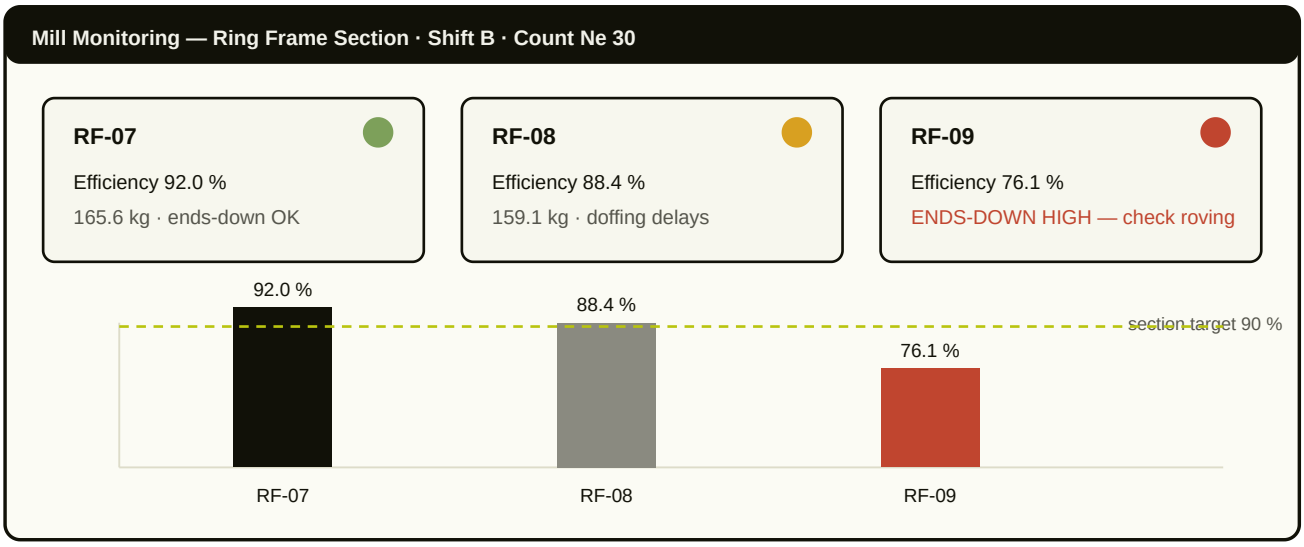


Figure 1 — A production-monitoring dashboard in schematic form: one card per machine with live efficiency and status, and the shift bar chart against the section target. The red card is the shift's management problem, identified in seconds.

3. Lab procedure

1. Set up the section in the software (or the lab's spreadsheet model of it): three ring frames of 1,000 spindles, count Ne 30, spindle speed 15,000 rpm, twist 20 TPI, shift length 480 min.
2. Enter the shift's actual production from the production register: RF-07 = 165.6 kg, RF-08 = 159.1 kg, RF-09 = 137.0 kg.
3. Compute the theoretical production (worked example) and let the system derive each frame's efficiency. Check the system's numbers against your hand calculation.
4. Enter the ends-down records. Set the alert threshold at 12 breaks per 1,000 spindle-hours and note which frames the system flags.
5. Assign downtime reason codes to RF-09's lost time from the supervisor's log, and generate the one-page shift report: production, efficiency ranking, exceptions, reasons.
6. Write the action line the report implies for the next shift (which frame, which check, whose job).

4. Worked example — theoretical production and efficiency

Front-roller delivery follows from spindle speed and twist: each spindle revolution inserts one turn, so the yarn delivered per minute is spindle rpm ÷ twist.

Delivery = 15,000 ÷ 20 = 750 in/min = 750 × 0.0254 = **19.05 m/min per spindle**

Per shift = 19.05 × 480 = 9,144 m/spindle → 1,000 spindles = 9,144,000 m

Ne 30 corresponds to tex = 590.5 ÷ 30 = 19.68 tex, i.e. 19.68 g per 1,000 m:

Theoretical production = 9,144,000 × 19.68 ÷ 1,000 g = **180.0 kg per shift**

FRAME	ACTUAL (KG)	THEORETICAL (KG)	EFFICIENCY	SYSTEM FLAG
RF-07	165.6	180.0	165.6 ÷ 180.0 = **92.0 %**	—
RF-08	159.1	180.0	**88.4 %**	Below 90 % target
RF-09	137.0	180.0	**76.1 %**	Below target + ends-down

Ends-down: the section ran 1,000 × 8 = 8,000 spindle-hours per frame. At the 12-per-1,000 alert threshold, a frame may record up to 8 × 12 = 96 breaks before flagging. RF-09 recorded 120 breaks → 120 ÷ 8 = **15.0 per 1,000 spindle-hours** — flagged. Because only RF-09 is high while its neighbours on the same roving supply are normal, the report points at the frame itself (rollers, travellers, spindle condition) before blaming the material — the standard first reading of a monitoring exception.

5. Practice tasks

1. A 500-spindle frame spins Ne 20 at 14,000 rpm with 18 TPI for a 480-minute shift. Its actual production is 128.8 kg. Calculate the theoretical production and the efficiency. (tex = 590.5 ÷ Ne.)
2. An 800-spindle frame records 96 ends-down in an 8-hour shift. Calculate the ends-down rate per 1,000 spindle-hours and state whether it crosses the 12-per-1,000 threshold.
3. All six frames of a section drop in efficiency on the same shift while ends-down rises everywhere. State why the monitoring pattern points away from individual machines, and name the two checks it suggests first.

Answer key

1. Delivery = 14,000 ÷ 18 = 777.8 in/min = 19.76 m/min. Per spindle per shift = 19.76 × 480 = 9,484 m. tex = 590.5 ÷ 20 = 29.53. Theoretical = 500 × 9,484 × 29.53 ÷ 1,000 g = **140.0 kg**. Efficiency = 128.8 ÷ 140.0 × 100 = **92.0 %**.
2. Spindle-hours = 800 × 8 = 6,400. Rate = 96 ÷ 6,400 × 1,000 = **15.0 breaks per 1,000 spindle-hours** — above the 12 threshold; the frame is flagged for investigation.

3. A fault that strikes every frame at once is a **common cause**, not six simultaneous machine failures: the shared inputs are the material and the atmosphere. First checks: (i) the **roving/fibre lot** changed at that shift (short fibre or high trash raises breaks everywhere); (ii) the **humidification** — a fall in relative humidity dries the fibre, raising static and breaks across the whole shed. Machine-level causes are investigated only after the shared inputs are cleared.

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